

chunyuac_HW3

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1.

Let m_1 and m_2 be the sample first and second moments, respectively, that is

$$m_1 = \sum_{i=1}^n X_i/n,$$
$$m_2 = \sum_{i=1}^n X_i^2/n.$$

Let $X \sim \text{Gamma}(\alpha, \beta)$. From the Table of Distribution Facts we know $E(X) = \alpha/\beta$, $V(X) = E(X^2) - E(X)^2 = \alpha/\beta^2$ and hence

$$E(X^2) = \frac{\alpha^2 + \alpha}{\beta^2}.$$

Now matching the first two moments we get

$$m_1 = \frac{\alpha}{\beta},$$
$$m_2 = \frac{\alpha^2 + \alpha}{\beta^2}.$$

and thus $m_2 - m_1^2 = \alpha/\beta^2$. So

$$\beta = \frac{\frac{\alpha}{\beta}}{\frac{\alpha}{\beta^2}} = \frac{m_1}{m_2 - m_1^2},$$

which is the desired estimator for β . Plug this back in to $m_1 = \frac{\alpha}{\beta}$ to obtain the estimator for α

$$\alpha = \frac{m_1^2}{m_2 - m_1^2}.$$

2.

(a)

Since $n \geq 30$, the 95% confidence interval is

$$\left[\bar{x} - z_{.025} \left(\frac{s}{\sqrt{n}} \right), \bar{x} + z_{.025} \left(\frac{s}{\sqrt{n}} \right) \right],$$

where $z_{.025}$ is such that $P(Z > z_{.025}) = 2.5\%$ when Z has the standard normal distribution, whose value can be found to be -1.96 in python as follows:

```
[1]: from scipy.stats import norm  
  
norm.ppf(1 - 0.025)
```

```
[1]: 1.959963984540054
```

Thus, numerically the confidence interval can be evaluated as follows:

```
[2]: import numpy as np  
  
x = 14.3  
s = 4.2  
n = 40  
z = norm.ppf(1 - 0.025)  
  
(x-z*s/np.sqrt(n), x+z*s/np.sqrt(n))
```

```
[2]: (12.998430432160422, 15.60156956783958)
```

(b)

Since n is small and the population distribution is unknown, none of the confidence intervals covered in the lecture is applicable. If σ were known, there is Chebyshev's inequality to guarantee

$$P\left(\bar{x} - k \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + k \frac{\sigma}{\sqrt{n}}\right) > 1 - \frac{1}{k^2}.$$

Setting $k = \sqrt{20}$ leads to a 95% confidence interval. But here σ is also unknown. The only clue is the sample standard deviation $s = 4.2$, which is not guaranteed to be close to σ either. There might still be a rigorous way to construct a 95% confidence interval but I feel like even if there is one, the constructed confidence interval will not be meaningful, maybe some wide range unless n is large, but if n is large we can just apply CLT.

(c)

Since $n < 30$ but the population distribution is known to be normal, the 95% confidence interval is

$$\left[\bar{x} - t_{.025,9} \left(\frac{s}{\sqrt{n}} \right), \bar{x} + t_{.025,9} \left(\frac{s}{\sqrt{n}} \right) \right],$$

where $t_{.025,9}$ is such that $P(T > t_{.025,9}) = 2.5\%$ when T has the t -distribution with 9 degrees of freedom, whose value can be found to be 2.262 in python as follows:

```
[3]: from scipy.stats import t  
  
t.ppf(1-0.025, 9)
```

```
[3]: 2.2621571627409915
```

Thus, numerically the confidence interval can be evaluated as follows:

```
[4]: x = 14.3  
s = 4.2  
n = 10  
p = t.ppf(1-0.025, 9)  
  
(x-p*s/np.sqrt(n), x+p*s/np.sqrt(n))
```

```
[4]: (11.295500994999198, 17.304499005000803)
```

3.

(a)

The log likelihood function is

$$l(\lambda) = \log \left(\prod_{i=1}^n \frac{e^{-\lambda} \lambda^{X_i}}{X_i!} \right) = \sum_{i=1}^n (-\lambda + X_i \log(\lambda) - \log(X_i!)).$$

Take the first derivative with respect to λ and set to zero to get

$$\frac{dl}{d\lambda} = \sum_{i=1}^n \left(-1 + \frac{X_i}{\lambda} \right) = -n + \frac{\sum_{i=1}^n X_i}{\lambda} = 0$$

and hence $\lambda = \sum_{i=1}^n X_i / n = \bar{X}$. Now check the second derivative

$$\frac{d^2 l}{d\lambda^2} = -\frac{\sum_{i=1}^n X_i}{\lambda^2} \leq 0.$$

Thus we can conclude that $\bar{X}$ is indeed the maximum likelihood estimator for λ .

(b)

Since $E(\bar{X}) = \sum_{i=1}^n E(X_i)/n = \lambda$, we know $\bar{X}$ is an unbiased estimator of λ .

(c)

The standard error can be found as

$$\sqrt{V(\bar{X})} = \sqrt{V\left(\frac{\sum_{i=1}^n X_i}{n}\right)} = \sqrt{\frac{\sum_{i=1}^n V(X_i)}{n^2}} = \sqrt{\frac{V(X_i)}{n}} = \sqrt{\frac{\lambda}{n}}.$$

4.

Scipy only generates exponential random numbers with parameter 1. To generate exponential random numbers with parameter λ we use this property: If $X \sim \text{Exponential}(1)$ then $X/\lambda \sim \text{Exponential}(\lambda)$.

As the result shown below, only 55% of the time the adjusted estimator actually generates a better estimate than the “worst” estimator does.

```
[5]: from scipy.stats import expon

n = 20
l = 10

numRepetition = 100000
numAdjBetter = 0

for _ in range(numRepetition):
    sample = expon.rvs(size=n)/l

    lMom2 = np.sqrt(2*n/((sample**2).sum()))
    lMom1 = 1/sample.mean()
    lAdj = lMom1*(n-1)/n

    if abs(lAdj - l) < abs(lMom2 - l):
        numAdjBetter += 1

numAdjBetter/numRepetition
```

```
[5]: 0.55311
```

5.

(a)

Since $E(\hat{\theta}_1) = \frac{4}{3}E(\bar{X}) = \frac{4}{3}E(X) = \theta$, we know that $\hat{\theta}_1$ is an unbiased estimator for θ .

(b)

We have

$$\begin{aligned}
 \text{bias}(\hat{\theta}_2) &= E(\hat{\theta}_2) - \theta \\
 &= \int_0^\theta x f_{X_{(n)}}(x) dx - \theta \\
 &= \int_0^\theta \frac{3nx^{3n}}{\theta^{3n}} dx - \theta \\
 &= \frac{3n\theta^{3n+1}}{(3n+1)\theta^{3n}} - \theta \\
 &= -\frac{\theta}{3n+1} \neq 0.
 \end{aligned}$$

Thus $\hat{\theta}_2$ is biased.

(c)

First we find MSE for both estimators:

$$\begin{aligned}
 \text{MSE}(\hat{\theta}_2) &= V(\hat{\theta}_2) + \text{bias}^2(\hat{\theta}_2) \\
 &= \left(\frac{3n}{(3n+2)(3n+1)^2} + \frac{1}{(3n+1)^2} \right) \theta^2 \\
 &= \frac{2\theta^2}{(3n+1)(3n+2)}, \\
 \text{MSE}(\hat{\theta}_1) &= V(\hat{\theta}_1) = \frac{16}{9}V(\bar{X}) = \frac{16}{9n} \frac{3}{80} \theta^2 = \frac{\theta^2}{15n}.
 \end{aligned}$$

Thus $\text{MSE}(\hat{\theta}_2) < \text{MSE}(\hat{\theta}_1)$ if and only if

$$\frac{2}{(3n+1)(3n+2)} < \frac{1}{15n}.$$

Since $n > 0$, this is equivalent to $30n < (3n+1)(3n+2)$, which solves to $\{n > 2.23385\} \cup \{n < 0.09948\}$. Since n can only be integers, this is the same as $n > 2$.